KAIZEN MATHS

Lesson starter

Proof by Induction

Induction with Sums

Name _____

S1

Prove each sum formula by mathematical induction.

Prove by induction that $\sum_{r=1}^{n} 4r - 2 = n(4(n + 1) - 4)$ 2 for $n \geq 1$.

S2

Prove each odd-number sum formula by mathematical induction.

Prove by induction that $\sum_{r=1}^{n} 2r + 5 = n(n + 3)$ for $n \geq 1$.

S3

Prove the square-number sum formula by mathematical induction.

Prove by induction that $\sum_{r=1}^{n} r^2 = n(n + 1)(2n + 1)$ 6 for $n \geq 1$.

S4

Prove each geometric sum formula by mathematical induction.

Prove by induction that $\sum_{r=1}^{n} 3^{r-1} = 3^n - 1$ 2 for $n \geq 1$.

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Show your working

KAIZEN MATHS

Exit ticket

Proof by Induction

Induction with Sums

Name _____

EA1

Prove each sum formula by mathematical induction.

Prove by induction that n Σ r = 1 3r + 2 = n(3(n + 1) + 4) 2 for n ≥ 1.

EA2

Prove each odd-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 2r − 1 = n(n + 0) for n ≥ 1.

KAIZEN MATHS

Exit ticket

Proof by Induction

Induction with Sums

Name _____

EB1

Prove the square-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 $3r^2$ = 3n(n + 1)(2n + 1) 6 for n ≥ 1.

KAIZEN MATHS

Exit ticket

Proof by Induction

Induction with Sums

Name _____

EA1

Prove each sum formula by mathematical induction.

Prove by induction that n Σ r = 1 3r + 2 = n(3(n + 1) + 4) 2 for n ≥ 1.

EA2

Prove each odd-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 2r − 1 = n(n + 0) for n ≥ 1.

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KAIZEN MATHS

Exit ticket

Proof by Induction

Induction with Sums

Name _____

EB1

Prove the square-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 $3r^2$ = 3n(n + 1)(2n + 1) 6 for n ≥ 1.

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Teacher answers & working

Proof by Induction · Induction with Sums

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S1

Prove each sum formula by mathematical induction.

Prove by induction that n Σ r = 1 4r − 2 = n(4(n + 1) − 4) 2 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 4 − 2 = 2.

$\text{RHS} = \frac{1(4(2)-4)}{2} = \frac{4}{2} = 2.$

Assume true for n = k: kΣr = 14r − 2 = $\frac{k(4(k+1)-4)}{2}$.

For n = k + 1, add the next term 4(k + 1) − 2.

$\text{k+1}\Sigma\text{r} = 14\text{r} - 2 = \frac{k(4(k+1)-4)}{2} + 4(\text{k}+1) - 2$

$= \frac{k(4(k+1)-4)+2(4(k+1)-2)}{2}$

$= \frac{(k+1)(4(k+2)-4)}{2}.$

This is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

S2

Prove each odd-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 2r + 5 = n(n + 3) for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 7.

$\text{RHS} = 1(1 + 3) = 7.$

Assume true for n = k: kΣr = 12r + 5 = k(k + 3).

For n = k + 1, add the next term 2(k + 1) + 5.

$\text{k+1}\Sigma\text{r} = 12\text{r} + 5 = \text{k}(\text{k} + 3) + 2(\text{k} + 1) + 5$

$= \text{k}^2 + 3\text{k} + 2\text{k} + 7$

$= \text{k}^2 + 5\text{k} + 7$

$= (\text{k} + 1)(\text{k} + 4).$

This is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

Teacher answers & working

Proof by Induction · Induction with Sums

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S3

Prove the square-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 r^2 = n(n + 1)(2n + 1) 6 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 1 × 1^2 = 1.

$\text{RHS} = \frac{1\times1\times2\times3}{6} = 1.$

Assume true for n = k: kΣr = $1r^2$ = $\frac{k(k+1)(2k+1)}{6}$.

For n = k + 1, add the next term $1(k + 1)^2$.

$k+1\Sigma r = 1r^2 = \frac{k(k+1)(2k+1)}{6} + 1(k+1)^2$

$= \frac{1(k+1)[k(2k+1)+6(k+1)]}{6}$

$= \frac{1(k+1)(2k^2+7k+6)}{6}$

$= \frac{1(k+1)(k+2)(2k+3)}{6}.$

This is $\frac{1(k+1)((k+1)+1)(2(k+1)+1)}{6}$, which is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

S4

Prove each geometric sum formula by mathematical induction.

Prove by induction that n Σ r = 1 3^{r-1} = 3^n − 1 2 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 3^0 = 1.

$\text{RHS} = \frac{3^1-1}{2} = \frac{2}{2} = 1.$

Assume true for n = k: kΣr = 13^{r-1} = $\frac{3^k-1}{2}$.

For n = k + 1, add the next term 3^k.

$k+1\Sigma r = 13^{r-1} = \frac{3^k-1}{2} + 3^k$

$= \frac{3^k-1+3^k(2)}{2}$

$= \frac{3^{k+1}-1}{2}.$

This is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

Teacher answers & working

Proof by Induction · Induction with Sums

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EA1

Prove each sum formula by mathematical induction.

Prove by induction that n Σ r = 1 3r + 2 = n(3(n + 1) + 4) 2 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 3 + 2 = 5.

$\text{RHS} = \frac{1(3(2)+4)}{2} = \frac{10}{2} = 5.$

Assume true for n = k: kΣr = 13r + 2 = $\frac{k(3(k+1)+4)}{2}$.

For n = k + 1, add the next term 3(k + 1) + 2.

$k+1\Sigma r = 13r + 2 = \frac{k(3(k+1)+4)}{2} + 3(k + 1) + 2$

$= \frac{k(3(k+1)+4)+2(3(k+1)+2)}{2}$

$= \frac{(k+1)(3(k+2)+4)}{2}.$

This is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

EA2

Prove each odd-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 2r − 1 = n(n + 0) for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 1.

$\text{RHS} = 1(1 + 0) = 1.$

Assume true for n = k: kΣr = 12r − 1 = k(k + 0).

For n = k + 1, add the next term 2(k + 1) − 1.

$k+1\Sigma r = 12r - 1 = k(k + 0) + 2(k + 1) - 1$

$= k^2 + 0k + 2k + 1$

$= k^2 + 2k + 1$

$= (k + 1)(k + 1).$

This is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

Teacher answers & working

Proof by Induction · Induction with Sums

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EB1

Prove the square-number sum formula by mathematical induction.

Prove by induction that n Σ r = 1 $3r^2$ = 3n(n + 1)(2n + 1) 6 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 3 × 1^2 = 3.

$\mathrm{RHS} = \frac{3\times1\times2\times3}{6} = 3.$

Assume true for n = k: kΣr = $13r^2$ = $\frac{3k(k+1)(2k+1)}{6}$.

For n = k + 1, add the next term $3(k + 1)^2$.

$k+1\Sigma r = 13r^2 = \frac{3k(k+1)(2k+1)}{6} + 3(k + 1)^2$

$= \frac{3(k+1)[k(2k+1)+6(k+1)]}{6}$

$= \frac{3(k+1)(2k^2+7k+6)}{6}$

$= \frac{3(k+1)(k+2)(2k+3)}{6}.$

This is $\frac{3(k+1)((k+1)+1)(2(k+1)+1)}{6}$, which is the required formula with n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.